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SUPERSEDING
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FEDERAL SPECIFICATION

MOWERS; LAWN, POWER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers lawn mowers with gasoline-engine power-driven cutting devices, which are self-propelled except for the size A, type II rotary, flat-knife type which is either self-propelled or hand-pushed. Gang cutting units as used herein include additional rotating reel assemblies either for installation on, or manual attachment to, the lawn mower. This specification also covers sulkies, weed cutters and grass catchers for attachment to the appropriate types of lawn mowers. This specification does not cover mowers with such service requirements or cutting capacities that the use of a separate power tractor or modified commercial truck is required.

1.1.1 *Federal specification coverage.* — Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications or which are commercially available, but are intended to cover the types, etc., which are ordinarily used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, styles, grades, and sizes.* —The power mowers covered by this specification shall be furnished in single or gang cutting units and in the following types, classes, styles, grades and sizes, as specified herein and in the invitation for bids, contract or order (see 6.1):

1.2.1.1 *Single cutting units.*

Type I.—Rotating-reel type.

Class A.—Roller drive.

Grade A. — Equipped with differential gearing.

Grade B. — Equipped with differential action.

Class B.—Wheel drive.

Grade A. — Equipped with differential gearing.

Grade B. — Equipped with differential action.

Sizes.—Type I, classes A and B mowers shall be furnished in the following sizes, as specified (see 6.1):

Size A.—18- to 21-inch cut.[1]

Size B.—22- to 24-inch cut.

Size C.—25- to 28-inch cut.

Size D.—29- to 32-inch cut.

[1] Type I, size A, lawn mower with wheel drive is generally available in grade B.

Type II.—Rotary flat-knife type (discs or arms rotating in a horizontal plane).

Style 1.—Hand-pushed.

Style 2.—Self-propelled.

Sizes.—Type II, styles 1 and 2 mowers shall be furnished in the following sizes, as specified (see 6.1):

Style 1—Hand-pushed

Size A.—19- to 21-inch cut.

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Style 2—Self-propelled

Size A.—19- to 21-inch cut.
Size B.—22- to 26-inch cut.
Size C.—27- to 31-inch cut.
Size D.—32- to 48-inch cut.
Size E.—50- to 60-inch cut.

Type III.—Sickle-bar type.
Size A.—36-inch cut (minimum).

Type IV.—Combination type (interchangeable rotating-reel and sickle-bar).
Size A.—30-inch cut (minimum).

1.2.1.2 *Gang cutting units.*

Type V.—Power-operated.
Group A.—For use with rotating-reel type mower, 29- to 32-inch cut, roller drive (type I, class A, size D).
Size A.—64-inch combined cut of lawn mower and power-operated gang.

Type VI.—Traction-wheel-operated.
Group A.—For use with rotating-reel type, 29- to 32-inch cut, roller drive or wheel drive (type I, classes A or B, size D).
Size A.—60-inch combined cut of lawn mower and traction-wheel-operated gang.

Group B.—For use with combination type (type IV mower), wheel drive.
Size A.—60-inch combined cut of lawn mower and traction-wheel-operated gang.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 There are no other specifications, standards, or publications applicable to this specification except those applicable only to individual departments listed in section 7.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

3. REQUIREMENTS

3.1 Condition.—Lawn mowers, gang cutting units, and attachment units shall be complete, new, and the latest improved models of the manufacturer. There shall be no defects or noticeable wear in any of the component parts of mechanisms upon the conclusion of any tests conducted to determine compliance with the service, cutting, or other performance requirements of this specification.

3.1.1 *Standard product.* — The equipment shall be essentially the standard product of the manufacturer so that prompt and continuing service and delivery of the spare parts may be assured. The component parts of the unit need not be the products of the same manufacturer.

3.2 Material. — Materials used in the construction of lawn mowers, gang cutting units and attachment units, shall be of the highest grade.

3.3 Construction.—Except for type II, style 1, lawn mowers shall be self-propelled. Unless otherwise specified, the cutting devices of the types of lawn mowers specified herein shall be gasoline-engine power-operated. Frames and structural supporting members shall be of either heavy or reinforced construction, and shall be capable of withstanding loads arising from operation on rough grounds without breakage or visible deformation. Component and related parts shall be accurately and permanently aligned. Assembled machines shall have adequate rigidity and shall be free from excessive vibration when in operation. All bearing housings shall adequately protect the bearings against the entrance of grass clippings, dirt, and other foreign material.

3.3.1 *Machine work.*—Tolerances and gages for metal fits shall conform to the limitations specified herein and otherwise to the standards of good commercial practice. Finished contact bearing surfaces shall be true and exact.

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3.3.2 *Guards.*—Effective guards shall be provided to overcome hazards created by reciprocating, rotating or moving parts such as chains, belts, sprockets, gears, etc., that are exposed to contact by the operator or others.

3.4 Service requirements.

3.4.1 *Machine duty.*—Lawn mowers shall be capable of operating continuously for eight hours daily, except for intervals required for refueling, under the cutting conditions and at the operating speeds hereinafter specified.

3.4.2 *Operation on level grounds.*—All lawn mowers shall be capable of continuous cutting at a speed of 3 miles per hour on firm and fairly level grounds or lawns. This requirement shall also apply to operation with specified gang cutting units and other specified attachment units, including operation with a mounted operator if a sulky is specified.

3.4.3 *Operation on grades.*

3.4.3.1 *Types I, III, and IV, lawn mowers.*—Types I, III, and IV mowers shall be capable of continuous cutting directly up firm grades of 30 percent at a speed of 2½ miles per hour. This requirement shall also apply to operation with specified gang cutting units and with other specified attachment units except the sulky. Sulkies, if furnished, shall be detached when lawn mowers are tested for operation on grades.

3.4.3.2 *Type II lawn mowers.*—Type II mowers shall be capable of continuous cutting directly up firm grades of 20 percent at a speed of 2½ miles per hour. Sulkies, if furnished, shall be detached when lawn mowers are tested for operation on grades.

3.4.4 *Adjustable operating speeds.*—The cutting speed on level lawn and on grades shall be readily varied over a range of speed which shall include 2 miles per hour. All cutting speeds specified herein, and any additional speeds at which the mower may be designed to operate continuously, shall be controlled either by a throttle, an automatic or manually adjustable speed governor, gear-shift transmission or a combination thereof.

3.5 Balance.—Lawn mowers shall not tip over of their own accord when placed in either direction of travel across a 100 percent grade (45° slope).

3.6 Engine.—The engine shall be either single, twin or multiple cylinder, and shall operate in two or four cycles, using gasoline as fuel. It shall have adequate reserve power to operate the lawn mower satisfactorily under intermittent overload conditions and to insure long service life. It may be either air-cooled or water-cooled. Air-cooled engines shall be so mounted that the air passages about the cylinder(s) will not become clogged with cut grass or clippings thrown by the cutting device. Water-cooled engines shall be equipped with efficient automobile-type radiator and power-driven fan. The radiator shall be so mounted that the air passages will not become clogged with grass clippings thrown by the cutting device. The engine shall not stall, excessively overheat, or unduly smoke, from the exhaust or crankcase during test, either for machine duty (specified in 4.2.3), or for engine cooling (specified in 4.4). Exhaust valves and valve seats for engines of over 1¾ horsepower equipped with float feed carburetors, shall be hard-faced. The starter shall be either of the hand-crank, hand-lever, or recoil type and shall be automatically released when the engine starts. The engine crankshaft shall be enclosed in an oil-tight housing and supported on either sleeve, ball or roller bearings of ample size. Four and two cycle engines shall have suitable method of lubrication to permit satisfactory operation under the requirements of service specified in 3.4 The fuel tank shall have capacity of not less than that required for two hours of operation of the mower, except that for type I mowers where a mixing-type valve of carburetor is used, a tank with a capacity of one quart or more will be accepted.

3.6.1 *Fuel system.*—Unless otherwise specified by the procuring agency, the engine shall be equipped with: (1) either the float-type carburetor or mixing-valve type of fuel system, (2)

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an oil-bath type air cleaner, which shall be removable for cleaning and (3) an efficient exhaust muffler or silencer (see 6.1).

3.6.2 *Ignition system.*—Unless otherwise specified by the procuring agency, the engine shall be equipped with an ignition system that shall have a high-tension moisture-resistant magneto. Magneto points shall be protected against interference by, or accumulation of, grass clippings, dirt, particles, and oil.

3.6.3 *Governor.*—Unless otherwise specified by the procuring agency, the engine shall be equipped with either an adjustable air vane or mechanical type of governor for controlling engine speed on the smaller and larger sizes, respectively.

3.7 Clutches.—Clutches shall be substantially constructed throughout. They shall have adequate durability and load capacity when the mower is being operated under conditions complying with the service and cutting requirements specified herein. The clutch lining material in clutches having lined faces shall be replaceable. Clutch adjustment shall be provided to compensate for slight slippage or weakening of the spring.

3.8 Controls. — Handle-grip controls and lever controls shall be positive in action and except for reverse position of the tractor control, shall remain locked or fixed in any required position under all operating and handling conditions of the mower. Unless otherwise specified on type II self-propelled mowers and on type I mowers over 21-inch, provision shall be made for disengagement of reel, knives or other cutting mechanism while moving mower from one area to another under its own power.

3.9 V-belt drives.—Where a V-belt drive is used, a movable shaft and pulley or other equally suitable device may be furnished to function as a clutch in engaging and disengaging the V-belt drive. V-belt drive shall be designed with either a take-up on the pulley or with an idler.

3.10 Handle frame.—The handle frame on lawn mowers larger than 21-inch cutting swath shall be of metal construction. Handle frames shall be sufficiently rigid to prevent whipping action or noticeable deformation when the mower is manipulated on turns or hills. Braces shall be added if required to strengthen and increase the rigidity of the handle.

3.11 Lubrication.—Gears and bearings shall be accessible for lubrication, except ball or roller bearings which are packed in grease and sealed. Lubricating points shall have grease fittings of the grease-gun injection type. The mower shall be lubricated when delivered.

3.11.1 *Lubricants.*—The lubricants shall be as recommended by the mower manufacturer and approved by the contracting officer.

3.12 Sulky.—When specified in the invitation for bids, riding sulkies shall be furnished with lawn mowers. They shall be of a design suitable for use with the type of mower furnished and shall be readily attached and detached. The sulky chassis shall be steel and shall be substantially constructed throughout. It shall have a durable saddle seat. Sulky wheels shall have solid, pneumatic or semipneumatic rubber tires, as specified in the invitation for bids. If the type of tire is not specified in the invitation for bids (see 6.1), semipneumatic rubber tires shall be furnished. Solid tires shall be permanently molded on metal rims. Pneumatic rubber tires shall have a sectional width of not less than 3 inches. The drawbar shall be steel and shall have an adjustble foot rest.

3.13. Finish.—Power lawn mowers and all attachment units shall be finished with a durable weatherproof paint or lacquer.

3.14 Instruction book.—The contractor shall furnish with each lawn mower either a booklet or pamphlet giving complete instructions for the operation, lubrication, adjustment, and care of the engine, mower, and attachment units. A pamphlet listing repair parts shall also be furnished with each lawn mower.

3.15 Tools.—A kit of tools shall be furnished with each power mower of over 21-inch cut. The kit shall contain all tools necessary for

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the proper care and maintenance of the mower and attachment units. An efficient grease gun shall be furnished with each lawn mower having grease cups of the grease-gun injection type.

3.16 Type I, rotating reel.—Type I, rotating-reel mower shall have a cylindrical rotating reel and a fixed bar knife. The ends of the cutting reel shaft shall be supported at the side frames. The side frames shall either directly or indirectly form the main structural supports for the mounting members of the engine, power drive, and traction drive. For mowers with readily detachable cutting reels, see type IV, combination type, (see 3.16.4).

3.16.1 *Cutting requirements.*—Rotating-reel type mowers shall be capable of cutting heavy, thick-growth grass, 3 inches in height above the bar knife, in a single pass when operating under conditions specified in 3.4, 3.4.1, 3.4.2, and 3.4.3.1. The cut shall be smooth and even for all heights of cut above the ground to which the bar knife is adjusted or elevated. The mowers shall not slip at the wheels or rollers nor mar the turf in normal operation.

3.16.1.1 *Cutting swath.*—The width of the cutting swath (actual width of cut) of sizes A, B, C, and D, shall be not less than 18, 22, 25, and 29 inches, respectively.

3.16.1.2 *Height of cut above ground.*—Unless otherwise specified, the height of cut shall be adjustable from ¾ to 2 inches above the ground. The height of adjustment shall be firmly secured and shall not slip or get out of adjustment during the operation of the machine over rough terrain. The reel assembly shall not be tilted upon adjustment in a manner that will contribute to poor cutting alignment.

3.16.2 *Grades.*

3.16.2.1 *Grade A, equipped with differential gearing.*—Grade A mowers shall have a differential gearing of either the full bevel-gear type or the full spur-gear type in the traction drive. No pawls, ratchets, or movable pins shall be used in conjunction with the operation of the differential gearing. Differential gears shall mesh fully and smoothly. The differential gearing shall be completely housed in a grease or oil-filled case.

3.16.2.2 *Grade B, equipped with differential action.*—The traction drive on grade B mowers shall have either a pawl-and-ratchet type of differential action, independent drive-control clutches, or a differential gearing complying with the requirements for grade A lawn mowers. Lawn mowers with independent drive-control clutches shall have an independently-operated clutch in each drive wheel or in each end section of the drive roller.

3.16.3 *Traction drives.*

3.16.3.1 *Class A, roller drive.*—Class A mowers shall have a sectional metal drive roller, the over-all length of which shall be approximately equal to the width of the cutting swath. All sections of the roller shall be power driven. Rollers shall be all-metal or rubber-covered metal, as specified in the invitation for bids. If neither is specified, the contractor may furnish either all-metal or rubber covered metal rollers. Rubber-covered rollers shall have the rubber permanently molded to each section of the roller. Sufficient traction shall be obtainable with all-metal rollers to operate the lawn mower in conformity with the requirements specified in 3.4.3.1.

3.16.3.2 *Class B, wheel drive.*—The distance between the wheel-drive centers (or between pairs of drive wheels) on class B mowers shall be not less than 18 inches. Wheels shall be all-metal, rubber-covered metal, or they shall be fitted with pneumatic or semipneumatic rubber tires, as specified in the invitation for bids. If the type of wheels is not specified in the invitation for bids, the contractor may furnish either all-metal, rubber-covered metal, pneumatic or semipneumatic tired wheels. Rubber-covered metal wheels shall have the rubber facing either properly stretched over the wheels or securely bolted, or permanently molded to metal rims, unless otherwise specified (see 6.1). Pneumatic rubber tires on lawn mowers of over 21-inch cut shall be of the industrial type, not less than two-ply and not less than 3 inches in sectional width. Sufficient traction shall be obtainable with all

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metal wheels to operate the mower in conformity with the requirements specified in 3.4.3.1. Drive axles or drive wheels shall have either plain ball, or roller bearings. Plain bearings shall be high-grade bearing metal, with suitable means of lubricating, or self-lubricating bearings. Ball bearing shall be precision type and shall be packed with grease and sealed.

3.16.4 *Controls.*—Unless otherwise specified by the procuring agency, the contractor may furnish mowers with either two or three primary controls with such additional equipment as may be required for the satisfactory operation of the unit furnished. The controls furnished shall be as specified hereinafter for two or three primary controls.

3.16.4.1 *Three primary controls.*—The functions of the three primary controls (handle-grip controls and/or lever controls) shall be for *(a)* regulating the engine speed, *(b)* engaging and disengaging the traction drive, and *(c)* engaging and disengaging the cutting-reel drive. Each primary control shall operate independently of the others. The mowers furnished either as walking units or with riding sulkies shall have the primary controls conveniently located and within reach of the operator's normal position.

3.16.4.2 *Two primary controls.*—The mower with two primary controls shall consist of one control to simultaneously regulate the engagement and disengagement of the cutting reel and traction drive, and the other control to regulate the engine speed, (operation of the two controls by one rod or lever is acceptable). The engine speed and traction drive controls shall be within reach of the operator's normal position, but the cutting-reel drive feature of the control, if separate, may be at some other readily accessible location on the unit.

3.16.5 *Power drive.*—Power for driving the traction roller or traction wheels shall be transmitted from the engine either by means of sprockets and roller chains, gear shafts and gears, V-belts, or a combination thereof. Teeth of gears for mowers of over 21-inch cut shall be machine-cut. Sprocket teeth shall be either machine-cut or accurately cast to size for mowers of over 21-inch cut; for mowers of 21-inch cut and under they may be stamped. Transmission gears on machines having gear drive only shall be totally enclosed in an oil- or grease-filled housing. All drives shall be in conformance with recognized engineering practices or specific manufacturing recommendations. The V-belt and chain drives shall be in accordance with the recommendations of the V-belt and chain manufacturers. Upon request, data or calculations shall be furnished to justify the design of the power drive on the basis of established engineering practices.

3.16.6 *Cutting reel.*—The cutting-reel drive must slip or shall include a device which will automatically disengage the cutting reel, when an object becomes wedged between a reel knife and the bar knife. The cutting reel shall be of rugged and durable construction and shall be ground accurately cylindrical with respect to the shaft. Not less than two reel knives shall be in contact with the bar knife at all times. The number of reel knives, and the speed at which the reel revolves relative to the speed of travel of the mower, shall be such that not less than 9 cuts are made by the reel in a linear travel distance of one foot of the mower. Spiders shall be either malleable iron or steel. Each reel knife shall be attached securely to each spider by either riveting or welding. The cutting-reel knives shall be either alloy or high-carbon tool steel, the Rockwell C number of which shall be not less than C-27, but in all cases the bed knife shall be harder than the reel knives when measured within $\frac{5}{16}$-inch of the cutting edge after the reel is finish-ground, except for mowers where both reel and bed knives are tempered to a minimum of Rockwell C-42. The cutting-reel shaft shall be supported on sleeve bearings, roller or ball bearings. Sleeve bearings shall be high-grade bearing metal.

3.16.7 *Bar knife.*—The bar knife shall be either alloy or high-carbon tool steel, the Rockwell C number of which shall be not less than C-32, but as specified in 3.16.6, in all cases the bed knife shall be harder than the reel knife when measured within $\frac{5}{16}$-inch of the cutting edge after the knife is finish-ground. The portion

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of the surface of the backing bar to which the bar knife is attached shall be finished to a plane surface. Either the backing bar or the cutting reel shall have a screw mechanism for setting proper clearance between the cutting knives and the bar knife. The clearance setting mechanism shall remain securely locked and shall not creep or loosen under all operating conditions of the lawn mower. The screw mechanism for setting proper clearances should be easily accessible so that adjusting tools will not slip out of screw slots or wrenches off or screw heads.

Types V and VI, group A.

3.16.8 *Gang cutting units.*—Two gang cutting units shall be furnished with each lawn mower when gang cutting units are specified.

3.16.8.1 *General requirements.*—Gang cutting attachment units shall be of durable and rugged construction and shall be suitable for operation with the power mower furnished. When the mower with gang cutting unit makes a turn on level lawn or a grade, there shall be no patches of uncut grass remaining in the total cutting swath. The cutting reel shall be mounted on either roller bearings or precision-type ball bearings. The cutting-reel knives shall be either alloy or high-carbon tool steel, the Rockwell C number of which shall be not less than 27, and the bar knife shall be not less than 32, but in all cases the bed knife shall be harder than the reel knife when measured within $\frac{3}{16}$-inch of the cutting edge after the knife is finish-ground, except for mowers where both reel and bed knives are tempered to a minimum of Rockwell C 42. The knife backing surface of the backing bar shall be finished to a plane surface. Either the backing bar or the cutting reel shall have a screw mechanism for setting proper clearance between the cutting knives and the bar knife. The clearance setting mechanism shall remain securely locked and shall not creep or loosen under all operating conditions of the cutting units. The cutting frequency of the reel knives shall be approximately equal to that of the knives on the cutting reel of the power mower. Cutting-reel spiders shall be either malleable iron or steel. Each reel knife shall be attached securely to each spider by either riveting or welding. Gang units with follow-up rollers shall have either steel or malleable iron roller brackets. The adjustable height of cut above the ground shall be equal to that of the power mower. The height of cut of delivered gang units shall be equal to that of the delivered mower. Lubrication shall comply with the requirements specified in 3.11.

3.16.8.1.1 *Type V, group A, power-operated type.*—The cutting reels shall be operated by power transmitted by suitable means from the lawn mower power drive. The units shall be adequately and properly supported, and if necessary, shall be suitably counterbalanced. The gang cutting units shall be either readily elevated, folded, or rotated about supporting pivots into a position either against the side frames or between the side frames, when operation is desired without the gang units or when the mower is traveling between locations. The total swath cut by the combination of lawn mower and gang units shall be not less than 64 inches.

3.16.8.1.2 *Type VI, group A, traction-wheel-operated type.*—The cutting reels shall be operated by the traction wheels of the unit. The total swath cut by the combination of lawn mower and gang units shall be not less than 60 inches. Traction-wheel type cutting units shall have suitable steel drawbars or pushbars for attachment to the lawn mower and shall be readily attached and detached. Each wheel shall have a gear drive. Gears and pinions shall be either cast iron or steel; the teeth shall be machine cut. The gear housing shall protect the driving means against the entrance of cut grass and dirt. Drive wheels or drive axles shall be mounted on either plain roller bearings, or nonadjustable, precision-type ball bearings. Ball bearings shall be packed in grease and sealed.

3.16.9 *Weed cutters.*—When specified in the invitation for bids, (see 6.1), rotating-reel type lawn mowers (type I) shall be furnished with power-operated weed cutters. They shall be suitable for the cutting of buckhorn, dandelion, and similar weeds ordinarily found growing on developed grass lawns. They shall not be required to cut heavy dense growths, nor thick, brittle-stem growths unsuited for final cutting with the rotating reel of the lawn mower. They shall be either of the rotating double or triple blade,

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rotating reel, or sickle bar types. The height to which the weeds are cut above the ground shall be within the cutting range of the lawn mower cutting reel. The weed cutter assembly shall be sturdily supported on the structural members of the machine, and shall be readily detachable. The weed cutters shall operate in front of, and simultaneously with, the lawn mower cutting reel. Rotating-blade and rotating-reel assemblies shall be mounted on either ball or roller bearings. Drive shafts employed to transmit power to sickle bars shall be mounted on either ball or roller bearings. The width of the cutting swath shall be not less than that of the lawn mower reel. The Rockwell C number of all cutting blades or knives shall be not less than 35.

3.16.10. *Grass catcher.*—When specified in the invitation for bids, type I, rotating-reel type lawn mowers (other than models equipped with rear wheel casters) shall be furnished with grass catchers. They shall be either of sheet metal, woven wire, or metal-and-canvas construction, unless otherwise specified in the invitation for bids. They shall be durable in service and shall be readily fastened to the lawn mower.

3.17 Type II, rotary flat-knife type.—Type II mowers shall have short cutting knives rigidly fastened horizontally either to the flat steel discs at the periphery, or to the ends of steel arm or arms. The cutting edges may be integral with the arms, at the option of the contractor. Discs and arms shall be attached to the ends of vertical drive shafts and shall rotate at proper speed in a plane parallel to the ground.

3.17.1 *Cutting requirements.*—The lawn mowers shall be capable of cutting cleanly, in a single pass, the hardiest of annual growths of buckhorn, crowfoot, dandelion, milkweed, thistle, tung grass, Bermuda, foxtail, and Johnson grass, wire grass, ragweed, foxgrass, and similar field growths when operating under the conditions specified in 3.4.1, 3.4.2, 3.4.3.2, and 3.4.4. They shall also be capable of cutting lawn grass on developed lawns in an even, uniform manner.

3.17.1.1 *Cutting swath.*—The width of the cutting swath (actual width of cut) of sizes A, B, C, D, and E shall be not less than 19, 23, 27, 32, and 50 inches, respectively.

3.17.1.2 *Height of cut above the ground.*—The height of cut above the ground shall be adjustable. The method of adjustment shall permit cutting to heights of from 1 inch to 3 inches above the ground. When specified, provision shall be made for adjustment of cutting heights from 2 to 4 inches and to prevent lowering the blade to cut below 2 inches.

3.17.2 *Differential gearing.*—Except for machines with single-swivel-wheel or front-wheel drive, the traction drive shall have a differential gearing of either the full bevel-gear type or the full spur-gear type. Differential gears shall mesh fully and smoothly. The differential gears shall be completely housed in a grease- or oil-filled case.

3.17.3 *Controls.*—Controls for engaging (and disengaging) the traction drive, and for regulating the engine speed shall be mounted on the handle frame within convenient reach of the operator. The engine stop switch may be mounted on the engine.

3.17.4 *Rotating discs and arms.*—Discs and arms shall be sufficiently heavy to prevent distortion or breakage when the mower is cutting heavy growths or operating on rough grounds. They shall be fastened to the vertical drive shafts with either machine or cap screws or bolts, and shall be readily detachable.

3.17.5 *Cutter blades.*—Cutter blades shall be either alloy or high-carbon tool steel, the Rockwell C number of which shall be not less than 35 at points adjacent to the cutting edge. They shall be securely riveted or bolted to the rotating discs or arms. If the cutting knives are integral with the arms, the arms shall be heat-treated steel, of spring temper, not less than 2¼ inches wide and for mowers with more than 21-inch cut, not less than ¼-inch thick, except at the sharpened cutting edges. The Rockwell C number at points adjacent to the cutting edges shall be not less than 35.

3.17.6 *Drive wheels.*—Drive wheels shall be of metal construction, and shall be fitted with

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rubber tires. Drive wheel tires shall be not less than two-ply, and not less than 4 inches in cross-sectional width, except for size A which may be 1½ inches. When guide wheels are used, the tires shall have a sectional width of not less than 2 inches.

3.17.7 *Guards.*—Adequate guards to protect personnel from the blades shall be provided. The blade housing, by the use of guards on all sides, shall provide adequate protection to the operator, and to others not operating the mower.

3.18 Type III, sickle-bar type.—The cutting device of type III lawn mowers shall consist of a power-operated sickle bar of the reciprocating cutter type.

3.18.1 *Cutting requirements.*—The machines shall be capable of cutting cleanly, in a single pass, the hardiest of annual growths of buckhorn, crowfoot, dandelion, milkweed, thistle, Bermuda, foxtail, Johnson grass, tung grass, wire grass, ragweed, foxgrass, and similar field growths when operating under the conditions specified in 3.4.1, 3.4.2, 3.4.3.1, and 3.4.4.

3.18.1.1 *Cutting swath.*—The width of the cutting swath (actual width of cut) shall be not less than 36 inches.

3.18.1.2 *Height of cut above the ground.*—A mechanical adjusting device shall be provided either on the sickle bar or on the machine to permit the sickle bar to be adjusted for heights of cut of from 1 inch to 3 inches above the ground.

3.18.2 *Sickle bar.*—The sickle bar shall be sturdily supported by the structural members of the machine. The drive mechanism shall be supported on roller bearings or precision-type, non-adjustable ball bearings. The finger bar shall be steel and shall be not less than 2 inches wide and $^{11}/_{32}$-inch thick. The guards shall be malleable iron, not less than 5½ inches long, and shall be individually bolted to the finger bar. The guards shall have steel serrated plates, which shall be securely riveted to the guards. Each finger bar shall have not less than four knife clips which shall be made of malleable iron and securely bolted to the bar. Wearing plates shall be fitted under the knife clips. The knife bar shall be steel and shall be fitted with blade sections not less than 2 inches wide. The blade sections shall be securely riveted to the knife bar. The Rockwell C number of the blade section at points adjacent to the sharpened edges shall be not less than 50.

3.18.3 *Drive wheels.*—Type III lawn mowers shall have not less than two drive wheels which, unless otherwise specified, shall be separated not less than 20 inches as measured between the wheel centers. Wheels shall be of metal construction and fitted with rubber tires.

3.18.4 *Power drive.*—Power for operating the sickle bar and the drive wheels shall be transmitted from the engine either by means of roller chains and sprockets, gears, and gear shafts, V-belts, or a combination thereof.

3.18.5 *Differential.*—Unless a particular type is specified (see 6.1), the traction drive shall have either a differential gearing, or each drive wheel shall have an independent drive control clutch. The differential gearing shall be either of the full bevel-gear type or the full spur-gear type, and shall be enclosed in a grease- or oil-filled housing.

3.19 Type IV, combination type.—Type IV lawn mowers shall be of such design and construction that either a rotating reel or a sickle bar can be readily attached to the machine and readily detached therefrom without disturbing the structural assembly of the machine. With the rotating reel attached the machine shall operate as a type I lawn mower; with the rotating reel removed and the sickle bar attached the machine shall operate as a type III lawn mower. The rotating reel and its bar knife shall be removable as an integral unit. Unless otherwise specified in the invitation for bids (see 6.1), type IV lawn mowers shall be furnished with both the rotating reel and the sickle bar.

3.19.1 *Cutting requirements.*

3.19.1.1 *With rotating reel attached.*—With the rotating reel attached the machine shall be capable of cutting heavy, thick-growth grass, 3

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inches in height above the bar knife in a single pass, when operating under the conditions specified in 3.4.1, 3.4.2, 3.4.3.1, and 3.4.4. The cut shall be smooth and even for all heights of cut above the ground to which the bar knife is adjusted or elevated. The mower shall not slip at the wheels or rollers, nor mar the turf.

3.19.1.2 *With sickle bar attached.* — With sickle bar attached the machine shall be capable of cutting cleanly, in a single pass, the hardiest of annual growths of buckhorn, crowfoot, dandelion, milkweed, thistle, Bermuda, foxtail, Johnson grass, tung grass, wire grass, ragweed, fox grass, and similar field growths when operating under conditions specified in 3.4.1, 3.4.2, 3.4.3.1, and 3.4.4.

3.19.1.3 *Cutting swath.* — The width of the cutting swath (actual width of cut) with either the sickle bar or the rotating reel shall be not less than 30 inches.

3.19.1.4 *Height of cut above the ground.*

3.19.1.4.1 *With rotating reel attached.* — With rotating reel attached the height of cut above the ground shall be adjustable over a range which shall include ¾ and 1½ inches; provided, however, that when this range of adjustment is obtainable only by use of separate gage shoes of different thicknesses, the height of cut with the gage shoes furnished shall be fixed at ¾ inch. The setting of height adjusting devices on delivered lawn mowers shall be such as to give a height of cut of ¾ inch. Devices for setting the cutting height shall not creep or loosen under all operating conditions of the lawn mower.

3.19.1.4.2 *With sickle bar attached.*—A mechanical adjusting device shall be provided to permit the sickle bar to be adjusted for heights of cut of from 1 inch to 3 inches above the ground.

3.19.1.5 *Cutting reel.*—The cutting reel shall be of rugged and durable construction, and shall be ground accurately cylindrical with respect to the shaft. Not less than two reel knives shall be in contact with the bar knife at all times. The number of reel knives, and the speed at which the reel revolves relative to the speed of travel of the mower, shall be such that not less than nine cuts are made by the reel in a linear travel distance of one foot of the mower. Spiders shall be either malleable iron or steel. Each reel knife shall be attached securely to each spider by either riveting or welding. The cutting reel knives shall be either alloy or high carbon tool steel, the Rockwell C number of which shall not be less than 35, but in all cases the bed knife shall be harder than the reel knives when measured within $\frac{3}{16}$-inch of the cutting edge after the reel is finish-ground except for mowers where both reel and bed knives are tempered to a minimum of Rockwell C 42. The cutting reel shaft shall be supported either on sleeve bearings, roller bearings, or precision-type ball bearings. Sleeve bearings shall be high-grade bearing metal.

3.19.1.6 *Bar knife.*—The bar knife shall be either alloy or high-carbon tool steel, the Rockwell C number of which shall be not less than 40, but as specified in 3.19.1.5, in all cases the bed knife shall be harder than the reel knife when measured within $\frac{3}{16}$-inch of the cutting edge after the knife is finish-ground except for mowers where both reel and bed knives are tempered to a minimum of Rockwell C 42. The portion of the surface of the backing bar to which the bar knife is attached shall be finished to a plane surface. Either the backing bar or the cutting reel shall have a screw mechanism for setting proper clearance between the cutting knives and the bar knife. The clearance setting mechanism shall remain securely locked and shall not creep or loosen under all operating conditions of the lawn mower.

3.19.1.7 *Sickle bar.*—The sickle bar shall be sturdily supported by the structural members of the machine. The drive mechanism shall be supported on roller bearings or precision-type, nonadjustable ball bearings. The finger bar shall be steel and shall be not less than 2 inches wide and $\frac{11}{32}$-inch thick. The guard shall be malleable iron not less than 5½ inches long and shall be individually bolted to the finger bar. The guards shall have steel serrated plates, which shall be securely riveted to the guards. Each finger bar shall have not less than four knife clips which

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shall be made of malleable iron and securely bolted to bar. Wearing plates shall be fitted under the knife clips. The knife bar shall be steel and shall be fitted with blade sections not less than 2 inches wide. The blade sections shall be securely riveted to the knife bar. The Rockwell C number of the blade sections at points adjacent to the sharpened edges shall be not less than 50.

Type VI, group B.

3.19.1.8 *Gang cutting units.* — Gang cutting units shall be furnished when specified in the invitation for bids (see 6.1), and shall comply with the requirements of 3.16.8, 3.16.8.1, 3.16.8.1.1 and 3.16.8.1.2.

3.19.1.9 *Drive wheels.*—Type IV lawn mowers shall have not less than two drive wheels which, unless otherwise specified, shall be separated not less than 20 inches as measured between the wheel centers. Wheels shall be of metal construction and shall be fitted with rubber tires.

3.19.1.10 *Differential.*—Unless a particular type is specified (see 6.1), the traction drive shall have either a differential gearing, or each drive wheel shall have an independent drive control clutch. The differential gearing shall be either of the full bevel-gear type or the full spur-gear type, and shall be enclosed in a grease- or oil-filled housing.

3.19.1.11 *Power drive.* — Power for operating the drive wheels and either the sickle bar or the rotating reel shall be transmitted from the engine either by means of roller chains and sprockets, gears and gear shafts, V-belts, or a combination thereof.

3.20 Workmanship. — Lawn mowers, gang cutting units, and attachment units shall be properly assembled and substantially constructed throughout. They shall be free from rust and imperfections which may affect the appearance or serviceability of the mower. Machined surfaces shall be accurately finished. Castings shall be smooth and free from blowholes, pits, foreign material or other defects.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Power lawn mowers and attachments shall be inspected and tested to determine general compliance with the requirements of this specification. Operating tests as hereinafter specified shall also be conducted to determine compliance with service requirements (see 3.4), the cutting requirements specified for each type (see section 3) and the requirements for proper functioning of operating parts. Grades and lawns or grounds shall be considered firm when the drive wheels or drive rollers leave no rut or distinctly visible depression due to the weight on them after passage of the operating lawn mower.

4.2 Operating test on level ground.—The operating test on level ground shall be conducted on a straight, fairly level course. The growth shall be representative of the type and/or height specified in the cutting requirements. The operator shall guide the equipment toward a landmark or a special marker slightly more than 500 feet distant. With the equipment operating at the highest speed attainable, or, if no sulky is specified, at a speed at which the operator can follow for the duration of the test, a cut shall be made for a time interval of not less than two minutes. The time interval shall be determined with either a stop watch or an accurate watch having a second hand. Only whole minutes shall be determined with the latter watch. The length of cut shall be measured with a tape measure. Equipment producing an unsatisfactory cut, but having traversed the course at a speed greatly in excess of 3 miles per hour, shall be retested for proper cutting at a reduced speed on a new course, but not at a speed of less than 3 m.p.h. As many additional cuts as may be necessary shall be made at a speed of 3 m.p.h. to determine compliance with the cutting requirements for the type of mower being tested.

4.2.1 *Calculation of speed.* — Speed shall be calculated as follows:

$$\text{Speed (miles per hour)} = \frac{\text{Distance (in feet)}}{88 \times \text{time (in minutes)}}$$

4.2.2 *Operating test on grades.* — The grade operating test shall be made between two marked points not less than 55 feet apart as measured along the specified grade. The grade shall be

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fairly uniform. The growth shall be representative of the type and/or height indicated in the cutting requirements. With the lawn mower operating at the highest speed attainable, or at the speed at which the operator can follow for the duration of the test, the time required for producing a cut between the two measured points shall be determined with a stop watch. The stop watch shall be started at the instant a marked spot on the lawn mower crosses the starting point and shall be stopped at the instant the same marked spot crosses the ending point. Lawn mowers producing an unsatisfactory cut, but having traversed the course at a speed greatly in excess of 2½ miles per hour, shall be retested for proper cutting at a reduced speed on a new course, but not at a speed of less than 2½ miles per hour. The ability of the lawn-mowing equipment to operate continuously on the grade at this speed shall be determined as part of the test for machine duty.

4.2.2.1 *Long course method (optional).*—If a stop watch is not available, the foregoing test may be conducted along the specified grade for an interval of time measured in whole minutes by use of the second hand of an accurate watch. The interval of time shall be either 2, 3, 4, or 5 minutes, depending on the length of uniform grade available. Care shall be exercised to see that the equipment is fully in motion at the highest speed at the start of the test and that forward motion of the equipment ceases at the end of the time interval. The distance traversed during the time interval shall be measured with a tape measure. The speed of operation shall be calculated as indicated in 4.2.1.1. (This method requires a test distance of 440 feet at 2½ m.p.h. for a time interval of 2 minutes.)

4.2.2.2 *Measurement of grades.* — Grades shall be measured with suitable instruments ordinarily employed for such purposes. When regular grade measuring instruments are not available, the grade shall be determined by use of a bench level, a protractor and a yardstick. The angle between the surface of the yardstick, when lying firmly on the grade, and the surface of the bench level, when leveled horizontally, shall be carefully measured with the protractor. The value obtained when the angle is subtracted from 180 shall be the measure of the grade in degrees. Grades of 20 and 30 percent are equivalent to inclines making angles of 11⅓° and 17°, respectively, with the horizontal.

4.2.2.3 *Calculation of speed.*—When a stop watch is employed and the time interval is determined in seconds, the speed shall be calculated by the following formula:

$$\text{Speed (miles per hour)} = \frac{\text{Distance (in feet)}}{1.46 \times \text{time (in seconds)}}$$

4.2.3 *Machine duty test.*—Lawn mowers with specified attachments shall be operated continuously over extensive lawns or grounds for a period which, in the estimation of the inspector, is sufficient to demonstrate that the equipment is capable of continuous and satisfactory operation. Where the purchase warrants, an area equivalent to ½ acre for each inch of rated swath width for the mower shall be mowed. Height of cut for the test of the rotating-reel types shall be 1½ inches and for the rotary flat-knife types shall be 2½ inches. Growth cut shall be heavy, thick-growth grass or grass and weeds, most of which has a minimum height of 3 inches above the specified level of cut. Operating speeds, equal to those specified in 3.4.2, 3.4.3.1, and 3.4.3.2, shall be maintained, both on level grounds and on the grades specified. If, in the judgment of the inspector, the equipment does not appear capable of maintaining the required cutting speeds, either during this test or upon its completion, he shall again conduct tests for cutting speeds over measured courses, both on level ground and on the grade specified. The ability of the equipment to operate satisfactorily at reduced speeds, especially at a speed of 2 m.p.h. shall be ascertained during the course of this test, unless it is found necessary to conduct additional tests similar to those specified in 4.2.1, and 4.2.2. Any unusual wear or failure of gears, pitman, pinions, clutches, frame, or other parts of the unit or failure to cut the grass in a uniform and satisfactory manner shall be cause for rejection.

4.3 Frequency of cut (types I and IV lawn mowers).—The frequency of cut of types I and IV mowers shall be calculated from the number of knives on the reel, the circumference of the

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drive wheels or rollers, and the ratio of the number of revolutions made by the reel to the number of complete revolutions made by the drive wheels or drive rollers. The ratio may be determined from the number of teeth in the gear or sprocket train, or by moving the mower without slipping the wheels or roller, or by turning the drive wheels or roller in any convenient manner. The frequency of cut shall be calculated as follows:

$$F = \frac{R \times N}{C}$$

Where:

F = Frequency (cuts per foot of travel).
R = Ratio of reel to wheel revolutions.
N = Number of reel knives.
C = Circumference of drive wheels.

4.4 Engine cooling test (optional). — The lawn mower without attachment units shall be driven over a fairly level concrete road for one hour with a drag attached. The drag shall consist of a wooden board of suitable size loaded with weights and fastened to the lawn mower by means of a rope or cable. The total weight of the drag shall be equal to one-half of the combined weight of the lawn mower and specified attachment units, plus 75 pounds if a riding sulky has been specified with the machine. The lawn mower shall be loaded with weights, if necessary, until sufficient traction is obtained. If necessary, either the lawn mower may be assisted to obtain its initial motion or the lawn mower and drag may be progressively loaded while in motion until the lawn mower exerts a tractive effort sufficient to drag the specified load. During the performance of this test, the cutting mechanism drive of rotating-reel type lawn mowers shall be disengaged to prevent possible heating of the bar knife and cutting blades. Power-driven gang cutting units need not be removed for performing this test. Stalling of the engine or excessive heat or smoke as prohibited by 3.6, shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.—Unless otherwise specified, the subject commodity shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to the point of delivery.

5.2 Marking.—Each item of equipment shall bear the manufacturer's name or trade-mark on a nameplate securely affixed in a conspicuous place. In lieu of the nameplate, the manufacturer's name or trade-mark may be cast integral with, stamped, or otherwise permanently marked upon the item of equipment. Unless otherwise specified, shipping containers shall be marked with the name of the material, the size, type, and the quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 Ordering data. — Purchasers should exercise any desired options offered herein and should specify the following information as specified:

(a) Title, number and date of this specification.

(b) Type, style, class, grade, and size (see section 1).

(For type I lawn mowers, the class, grade, size, and the type drive wheels or drive rollers should be specified.

For type II lawn mowers, the size and style, if applicable, should be specified).

(c) Options:

(1) Type of engine required (see 3.6).
(2) Type of fuel system required (see 3.6.1).
(3) Ignition system, if different from that specified (see 3.6.2).
(4) If governor is not required, so specify (see 3.6.3).
(5) Whether provision is desired for disengagement of reel, knives, or other cutting mechanism (see 3.8).

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(6) Whether sulkies are desired. If so, type of tires required (see 3.12).

(7) Type rollers required for class A mowers (see 3.16.3.1).

(8) Type of wheels desired for class B mowers (see 3.16.3.2).

(9) Number of controls desired (see 3.16.4).

(10) Specify whether weed cutters are desired (see 3.16.9).

(11) Specify grass catcher, if required (see 3.16.10).

(12) Specify whether differential gearing, or independent drive control clutch is desired (see 3.18.5 and 3.19.1.10).

(13) State whether both rotating reel and sickle bar are desired (see 3.19).

(14) Specify whether gang cutting units are required (see 3.19.1.8).

(15) Packaging, packing, and marking requirements (see section 5).

6.2 Definition of terms used in specification.—The term lawn mowers as used herein refers to hand-pushed, self-propelled machines (designated by types in 1.2.1), having power-driven cutting devices which produce cutting swaths of the sizes specified in 1.2.1. The term gang cutting units, as used herein, refers to additional rotating-reel assemblies which are either installed on the lawn mower or may be manually attached to the lawn mower. In order to meet the service and cutting requirements, competitive lawn mowers which are furnished with two gang cutting units, will be equipped with the largest size engines generally employed in lawn mower construction. In addition to the lawn mowers and the gang cutting units, this specification also covers sulkies (3.11), weed cutters (3.16.9) and grass catchers (3.16.10), which may be attached to the appropriate types of lawn mowers.

6.3 This specification does not apply to power lawn mowers which are pushed or propelled by manual effort, nor to lawn mowing equipment with such service requirements or cutting capacity that the use of either a separate power tractor or modified commercial truck is required. For hand lawn mowers, see Federal Specification OO–M–671.

6.4 Guarantee. — It is suggested that purchasing officers require a guarantee against defective material and/or workmanship for a period of not less than 90 days from date of delivery and that replacement be required of all parts found defective within the guarantee period, without cost to the Government.

6.5 Where is is deemed necessary, the contractor should be required to state the nearest source of supply of repair parts.

6.6 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.7 Transportation description.—Transportation description applicable to this item is:

Lawn mowers and engines combined. Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Lawn mowers and engine or motors, combined. Truckload minimum weight 24,000 pounds.

Minimum weights shown apply when material is transferred at classification ratings. They will vary when material moves on commodity rates. When in doubt consult General Services Administration Regional Traffic Manager.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that in any way be related thereto.

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7. DEPARTMENTAL REQUIREMENTS:

7.1 The following specifications, standards and other publications of the issues in effect on date of invitation for bids form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy and Air Force.

7.2.1 *Applicable specifications, standards and other publications.*

Federal Specifications:

UU–T–116—Tape; Paper, Gummed, Water-Resistant.

Military Specifications:

JAN–P–72—Primer, Synthetic, Lacquer-Resisting.

JAN–P–100—Packaging and Packing for Overseas Shipment — General Specification.

JAN–P–104—Packaging and Packing for Overseas Shipment — Crates, Sheathed, Wood, Nailed.

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed (Overseas Type).

JAN–P–115—Packaging and Packing for Overseas Shipment—Compound, Sealing, Dipcoating.

MIL–P–116—Preservation, Methods of.

JAN–P–117—Packaging and Packing for Overseas Shipment—Bags, Interior Packaging.

JAN–B–121—Barrier-Materials, Greaseproof.

JAN–P–125—Packaging and Packing for Overseas Shipment — Barrier-Materials, Waterproof, Flexible.

JAN–P–127—Packaging and Packing for Overseas Shipment — Tape, Adhesive, Pressure-Sensitive, Water-Resistant.

MIL–B–130—Barrier-Material, Paper, Non-corrosive.

MIL–L–2105—Lubricant, Gear, Universal.

MIL–C–3769 — Crates, Intermediate, Sheathed, Wood, Nailed (For Maximum Net Loads of 3000 Pounds).

MIL–E–11275—Engines, Gasoline, Industrial Type, ½ Thru 100 BHP, General Specifications.

MIL–P–15424—Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standard:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of specifications and standard required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Other publications.*

Consolidated Classification Committee Publication:

Consolidated Freight Classification Rules and Container Specifications.

American Trucking Association Publication:

National Motor Freight Classification Rules and Container Specifications. (Either No. 11 or No. A–1).

(Application for copies of the Consolidated Freight Classification Rules and Container Specifications should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.

(Application for copies of National Motor Freight Classification Rules and Container Specifications should be addressed to the American Trucking Association, 1424 16th Street, N.W., Washington, D. C.)

7.2.3 *Preservation and packaging.*

7.2.3.1 *Domestic shipment for immediate use.*—The mowers, component parts and attachments shall be preserved and packaged to afford protection against deterioration and damage.

7.2.3.2 *Domestic shipment involving limited storage.*—The mowers, component parts and attachments shall be preserved and packaged as specified herein for overseas or domestic shipment involving extended storage.

7.2.3.3 *Overseas shipment or domestic shipment involving extended storage.*

7.2.3.3.1 *Disassembly.*—Parts vulnerable to damage or that would otherwise increase cubage such as the controls, control rods, and levers shall

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be detached or removed from the handles and the handles removed from the mowers. Bolts, nuts, washers, and pins removed shall be placed and secured in one of mating parts to prevent their loss.

7.2.3.3.2 *Cleaning, drying and preservative.* —Surfaces to which a preservative is to be applied shall be cleaned and dried in conformance with Military Specification MIL–P–116. Preservative specified herein shall conform to the applicable specification listed in table I of Military Specification MIL–P–116.

7.2.3.3.3 *Unprotected metal surfaces.*—Unpainted or unprotected metal surfaces of the mowers, component parts, and attachments subject to corrosion such as cutting reels, cutter bars, sickle bar, sickle knives, bolts and nuts, shafts or any other unprotected metal surface not specifically specified herein shall be coated with type P–1, grade 1 preservative.

7.2.3.3.4 *Engines.*—The liquid-cooled or air-cooled engines shall be preserved and packaged in conformance with Military Specification MIL–E–11275.

7.2.3.3.5 *Sulky wheels.* — Wheels removed from sulkies to reduce cubage shall have the axles or spindles, the interior bearing surfaces of the hub, the exterior bearing surfaces of the axle or spindle, and all ball or roller bearings which are not removed from the axles or spindles coated with type P–6 preservative. The coated surfaces of the axles or spindles and the bearings shall be wrapped with greaseproof barrier-material type II, grade A, class 1, and overwrapped with greaseproof barrier-material, type II, grade C, class 1, both conforming to Military Specification JAN–B–121. The wrap shall be coated with dipcoating sealing compound conforming to Military Specification JAN–P–115. The wrap shall be protected by overwrapping with barrier-material conforming to Military Specification MIL–B–130, type I, secured in place with pressure-sensitive tape conforming to Military Specification JAN–P–127, type I, grade B.

7.2.3.3.6 *Drive wheels.*—Drive wheels shall be removed from the mowers and the interior surfaces, gears, and drive control clutches coated with lubricant conforming to Military Specification MIL–L–2105 and the wheels replaced.

7.2.3.3.7 *Chains and sprockets.* — Exposed drive chains shall be coated with enough type P–3 preservative to assure penetration to the inner surfaces of the rollers, pins, and bushings. After excess type P–3 preservative has drained, the entire chain shall be coated with type P–1, grade 1 preservative. Sprockets and other unprotected surfaces of the drives shall be coated with type P–1, grade 1, preservative.

7.2.3.3.8 *Drive belts and pulleys.* — Pulley faces or grooves shall be coated with lacquer-resisting synthetic primer conforming to Military Specification JAN–P–72. The drive belts shall remain in place on the equipment and shall have the tension relieved.

7.2.3.3.9 *Gears and gear housings.* — Gear housings shall be drained and flushed with petroleum solvent agitated with compressed air. The solvent shall be drained, the drain and filler plugs coated with type P–1, grade 1 preservative. The housing shall be filled to the operating level with lubricant conforming to Military Specification MIL–L–2105 and the gears operated to insure coating of all interior parts with the lubricant. Exposed gears shall be coated with type P–1, grade 1 preservative.

7.2.3.3.10 *Clutches.* — Metal components of clutches subject to corrosion shall be coated with lacquer-resisting synthetic primer conforming to Military Specification JAN–P–72.

7.2.3.3.11 *Instruction books.*—The instruction books shall be packaged in conformance with Military Specification MIL–P–116, either method IA–8, using interior packaging bags conforming to Military Specification JAN–P–117, type I, grade A, class d, with the bags heat-sealed, or method IA–1, and a preliminary wrap of kraft paper.

7.2.3.3.12 *Tools.* — The tools shall be preserved and packaged in conformance with Military Specification MIL–P–15424 and replaced within the kit. The kit shall be wrapped with

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waterproof barrier material conforming to Military Specification JAN–P–125, type C–1 or C–2, secured with pressure-sensitive tape conforming to Military Specification JAN–P–127, type I, grade B.

7.2.3.3.13 *Intermediate packaging.* — Components and attachments requiring additional protection shall be wrapped with waterproof barrier-material conforming to Military Specification JAN–P–125, type C–1 or C–2, secured with pressure-sensitive tape conforming to Military Specification JAN–P–127, type I, grade B.

7.2.4 *Packing.*

7.2.4.1 *Domestic shipment for immediate use.* — The mowers and attachments shall be packed in conformance with the requirements of Consolidated Freight Classification Rules and Container Specifications for rail shipments or with National Motor Freight Classification Rules and Container Specifications (either No. 11 or No. A–1) for truck shipments.

7.2.4.2 *Domestic shipment and storage or domestic reshipment.*—The mowers and attachments shall be packed as specified herein for overseas shipment.

7.2.4.3 *Overseas shipment.* — The mowers, packaged and unpackaged components, attachments, instruction books and operating tools comprising a complete unit not exceeding 200 pounds shall be packed in nailed wood boxes conforming to Military Specification JAN–P–106, style 2, 2½, or 3. When the gross weight exceeds 200 pounds and not more than 500 pounds, the nailed wood boxes shall be modified by the addition of not less than two nominal 2-inch by 4-inch skids attached and positioned across the width and flush with the outside of the box, spaced not to exceed 30 inches on centers. The ends of skids shall be beveled one-half the depth at 45 degrees. Cushioning and blocking shall be provided. The boxes shall be closed and strapped in conformance with the appendix to the box specification, except the straps on modified boxes shall be placed adjacent to the skids. When the gross weight exceeds 500 pounds, nailed wood sheathed intermediate crates conforming to Military Specification MIL–C–3769, with double-sheathed top shall be the containers. The crates shall be blocked, braced, and anchored in conformance with Military Specifications JAN–P–100 and JAN–P–104. Closure and strapping shall be in conformance with the crate specification.

7.2.5 *Performance.* — The packaging and packing of the mowers for overseas shipment accomplished in accordance with this specification shall be capable of withstanding one of the tests described in 7.2.7 without shifting, loose or broken hold-downs, ties, stays, blocking, or bracing.

7.2.6 *Pilot pack.*—Prior to packing the mowers in quantity, a pilot pack shall be submitted to determine conformance with the packing requirements of this specification, except the pilot pack will not be required when the vendor has previously had such a pack tested and accepted for identical equipment on a Military contract, and can furnish satisfactory evidence that the mowers are being packed in conformance with the approved pack. Approval of the pilot pack shall not relieve the contractor of his obligation to preserve, package, and pack the mowers in conformance with this specification.

7.2.7 *Pilot pack tests.*—Pilot pack tests shall apply only to crated shipment.

7.2.7.1 *Impact test.* — To determine if the pack is adequately blocked, braced, and anchored, and that provision has been made against shifting, one of the following tests shall be made. The railroad car method shall be used whenever such facilities are available. The crane and drop methods are alternate methods and shall be used in that order of preference only when facilities for the railroad car method are not available.

7.2.7.1.1 *Railroad car method.* — The pack shall be securely blocked to prevent movement on the car. Anti-skid plates shall not be used for the test. The car shall be humped to cause impact at approximately 10 miles per hour.

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7.2.7.1.2 *Crane method.*—The pack shall be raised with a crane and slings or grabhooks to a height sufficient to allow the pack to swing clear of the ground. The pack shall be swung to strike against a solid barrier at a speed of approximately 8 miles per hour. This can be done by pulling the pack back and up along the arc of free swing until the center of gravity reaches a point approximately 27 inches above its original position, also point of impact measured vertically. The test shall be repeated for the opposite end of the pack.

7.2.7.1.3 *Drop method.*—One end of the pack shall be mounted on a wood sill 5 inches or 6 inches high, placed at right angles to the skids. The opposite end shall then be raised to a height of 36 inches and allowed to fall freely to a hard surface. This test shall be applied once to each end of the pack.

7.2.8 *Examination, redesign, and retest.*—After the pilot pack test, the pack shall be examined for conformance with 7.2.5. Any deficiencies shall be corrected by redesign and the pack retested for conformance with this specification.

7.2.9 *Marking for shipment and storage.*—In addition to any special marking required by the contract or order, marking for identification, storage and shipment shall be in accordance with Military Standard MIL–STD–129. The approved item name shall be as follows: Mowers, Lawn, Power.

MILITARY INTEREST:

Army—E
Navy—Y S MC
Air Force.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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